Soomin Park

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RESEARCH INTEREST

My research interests lie in generative modeling for human motion, spanning both kinematic and physics-based approaches. I am currently developing humanoid control methods that enable flexible motion adaptation across diverse tasks and whole-body control with dexterous hands for precise object interaction. My research goal is to contribute to advancements and applications in computer animation, robotics, and AR/VR.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) **Daejeon, South Korea**
Ph.D. Candidate, Graduate School of Culture Technology *Mar. 2021 – present*
Lifelike Avatar & Agent Lab (Advisor: Sung-Hee Lee)

Korea Advanced Institute of Science and Technology (KAIST) **Daejeon, South Korea**
M.S., Graduate School of Culture Technology *Mar. 2019 – Feb. 2021*
Lifelike Avatar & Agent Lab (Advisor: Sung-Hee Lee)
- Thesis: "Spatio-Temporal Graph-Based Generative Adversarial Networks for Character Motion Style Transfer"
- Thesis committee: Sung-Hee Lee, Juhan Nam, Jaegul Choo

Ewha Womans University **Seoul, South Korea**
B.S. in Computer Science and Engineering *Mar. 2014 – Feb. 2019*

PROFESSIONAL EXPERIENCE

Roblox Corporation **San Mateo, CA, United States**
Research Intern (Host: Mubbasir Kapadia) *Aug. 2026 – Dec. 2026*
- Conducting research on generating character animation for human-object interaction (HOI) tasks.

MOVIN **Daejeon, South Korea**
Research Collaborator *Aug. 2024 – Feb. 2025*
- Implemented a pipeline to register motion capture data with LiDAR-based human point clouds.
- Mitigated drifting issues in IMU-based motion capture data.

Graphics Lab, Ewha Womans University **Seoul, South Korea**
Undergraduate Research Intern (Advisor: Young Jun Kim) *Jul. 2017 – Jun. 2018*
- Reduced rendering overhead for a high-resolution VR painting system.
- Demonstrated the system at KSC 2017 SW Implementation/Demo Contest.

PUBLICATIONS (*: EQUAL CONTRIBUTION)

Refereed Journals/Conferences

- *[4] "PhaseDance: Capturing Rhythm and Expressivity in Dance Modeling"*
  Meongeun Kim, Taehui Lee, **Soomin Park**, and Sung-Hee Lee
  - Conference on Neural Information Processing Systems (**NeurIPS**) 2026
- *[3] "MaskAdapt: Learning Flexible Motion Adaptation via Mask-Invariant Prior for Physics-Based Characters"*
  Soomin Park, Eunseong Lee, Kwang Bin Lee, and Sung-Hee Lee
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) 2026
  - Highlight paper (top 3.6% of submissions)

- *[2] "Motion Puzzle: Arbitrary Motion Style Transfer by Body Part"*
  Deok-Kyeong Jang, **Soomin Park**, and Sung-Hee Lee
  - ACM Transactions on Graphics (**TOG**), presented at **SIGGRAPH** 2022
- *[1] "Diverse Motion Stylization for Multiple Style Domains via Spatial-Temporal Graph-Based Generative Model"*
  Soomin Park, Deok-Kyeong Jang, and Sung-Hee Lee
  - Proceedings of the ACM on Computer Graphics and Interactive Techniques (**PACMCGIT**), presented at **SCA** 2021

Extended Abstracts, Posters, and Demos

- *[2] "Spatial Chef: A Spatial Transforming VR Game with Full Body Interaction"*
  Yeeun Shin*, Yewon Lee*, Sungbaek Kim*, and **Soomin Park***
  - CHI Conference on Human Factors in Computing Systems (**CHI EA**) 2023
- *[1] "CanvoX: High-resolution VR Painting in Large Volumetric Canvas"*
  Yeojin Kim, **Soomin Park**, and Young Jun Kim
  - KSC SW Implementation/Demo Contest 2017

RESEARCH PROJECTS

Development of Object Media Technology Based on Multiple Video Sources — **Daejeon, South Korea**
Institute for Information & Communications Technology Planning & Evaluation — *2021 – 2025*
- Led research on human motion generation and retargeting.
- Developed physics-based humanoid control frameworks for realistic motion synthesis.

Construction of Testbed and High-Fidelity Dataset for Pedestrian Perception Evaluation — **Daejeon, South Korea**
KATECH — *2025*
- Led development on a registration pipeline aligning LiDAR, GPS, and motion capture data.
- Constructed a dataset for evaluating pedestrian perception in autonomous driving.

Study of Styled Motion Generation for Non-Verbal Communication of Virtual Human Agents — **Daejeon, South Korea**
National Research Foundation of Korea — *2020 – 2022*
- Developed a style-controllable motion generation framework for humanoid agents.
- Developed evaluation metrics (FMD, SRA, CRA) for generative motion models.

HONORS AND AWARDS

- Highlight Paper, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- Student Game Competition Finalist, ACM CHI Conference on Human Factors in Computing Systems 2023
- Dean's Award for Academic Excellence (×4), Ewha Womans University 2014 – 2018
- Engineering Capstone Design Contest Participation Award, Ewha Womans University 2018
- Engineering Portfolio Contest Participation Award, Ewha Womans University 2018
- Excellence Award (3rd place), KSC 2017 SW Implementation/Demo Contest, Korea Software Congress 2017

ACADEMIC SERVICES

- Conference Reviewer
  : NeurIPS (2026), CVPR (2026), ECCV (2026), SIGGRAPH (2025), Pacific Graphics (2021)
- Journal Reviewer
  : TVCG (2025)

TECHNICAL SKILLS

- **Languages**: English (fluent), Korean (native)
- **Programming Languages**: Python, C/C++, C#, MATLAB
- **Frameworks & Libraries**: PyTorch, TensorFlow, NumPy, SciPy
- **Platforms & Tools**: Isaac Gym, Isaac Lab, MuJoCo, Bullet, Unity3D, Blender, Git, ROS
- **Core Skills**: Reinforcement Learning, Generative Models, Human Body/Motion Modeling, Data Analysis